SIMONE MANTI
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Personal contact details, birth data, photograph and signature are intentionally omitted in accordance with call no. 28978, Art. 7(h).

SCIENTIFIC PROFILE

Experimental and computational physicist working at the interface of laboratory X-ray spectroscopy, radiation detectors and data-driven modelling. Since 2022 at INFN-LNF, my research has combined Bragg-crystal X-ray emission and absorption spectroscopy, CZT and BEGe detector simulations and event selection, atomic-structure and cascade calculations, and first-principles high-throughput materials modelling. I am co-leader of the Software Advancement Working Group of COST Action ENRICH and an instructor in X-ray-optics ray tracing with SHADOW4/OASYS2.

WORK & RESEARCH EXPERIENCE

May 2022–present
Postdoctoral Researcher (Assegnista di ricerca), *INFN, Laboratori Nazionali di Frascati*
Laboratory X-ray spectroscopy, radiation-detector studies, precision atomic physics, simulation and machine-learning methods. Member of the VOXES, SIDDHARTA-2 and VIP research activities.

Sep. 2021–Apr. 2022
Research Assistant, *Technical University of Denmark (DTU)*
First-principles and high-throughput modelling of two-dimensional materials and quantum point defects. Supervisor: Prof. Kristian S. Thygesen.

EDUCATION

Awarded Nov. 2021
Ph.D. in Physics, *Technical University of Denmark (DTU)*
May 2018–Sep. 2021. Thesis: *Effects of lattice imperfections on the optical and electronic properties of two-dimensional materials.* Supervisor: Prof. Kristian S. Thygesen.

Awarded Dec. 2017
M.Sc. in Physics, 110/110 cum laude, *University of Rome Tor Vergata*
Apr. 2016–Dec. 2017. Thesis: *Real-time photoionization of atoms: successes and limitations of the non-equilibrium Green's-function method.* Supervisor: Prof. Gianluca Stefanucci.

Awarded Oct. 2015
B.Sc. in Materials Science, 104/110, *University of Rome Tor Vergata*
Oct. 2012–Oct. 2015. Thesis on the formation and characterization of defects on differently oriented germanium surfaces.

RESEARCH CONTRIBUTIONS RELEVANT TO PRISM

Laboratory X-ray spectroscopy and VOXES. I operate and optimize the INFN-LNF VOXES von Hamos spectrometer, including crystal/slit alignment, energy calibration, acquisition and background suppression. As first and corresponding author, I reported a one-hour multi-element Cu–Ni–Zn XES measurement with sub-10-eV FWHM resolution near 8 keV and an average 30% reduction in peak-centre uncertainty after background selection. A 2026 follow-up documents the ED-XRF monitor, liquid-sample handling and mechanical compatibility with laboratory transmission XAS.

MITIQO and quantitative XRF. I have responsibility for the implementation of MITIQO (project no. A0375-2020-36647), a Regione Lazio research project on trace metals and oxidation states in edible liquids. My contributions include WD- and ED-XRF measurements, experimental optimization, signal-to-background improvement, construction of elemental-composition datasets, principal-component analysis and machine-learning classification of wines by geographical origin.

Semiconductor detectors, Monte Carlo and data analysis. For CZT detectors at DAΦNE, I use detailed Geant4 simulations to generate labelled signal/background datasets and develop supervised event classifiers; this work was presented at XGRADE 2026. For the VIP experiment, I co-developed and validated machine-learning event selection for a BEGe detector (co-first and corresponding author, JINST 2026). These activities cover detector response, spectral and waveform features, pile-up/noise rejection and quantitative performance assessment.

Precision atomic and quantum X-ray spectroscopy. Within SIDDHARTA-2, I perform multi-configuration Dirac–Fock calculations and develop atomic-cascade simulations for kaonic atoms, connecting calculated transition energies and yields to measurements. I am first and corresponding author of the 2026 Physical Review A test of bound-state QED with kaonic neon. Within VIP, I contribute to first-principles transition calculations and X-ray tests of the Pauli Exclusion Principle and dynamical-collapse models; I am a corresponding author of the related 2024

Physical Review Letters article.

First-principles materials modelling and high-throughput workflows. My doctoral research used density-functional theory, automated workflows and databases to predict structural, electronic and optical properties of 2D materials. Contributions include dynamical-stability classification, the C2DB ecosystem and the QPOD database of more than 1,900 defect systems. This work produced first/corresponding-author and equal-contribution publications in *npj Computational Materials*, together with an article in *ACS Nano*.

RESPONSIBILITIES AND RECOGNITION

2025–present

Co-Leader, WG3 Software Advancement, COST Action CA24131 ENRICH

Coordinating activities on software development and standards for the European radiation-detection research network; WG leader: Dr Manuel Sanchez del Rio.

2026

Instructor, ESRF–ENRICH OASYS School EOS26

Hands-on training in X-ray-optics design and ray tracing with SHADOW4/OASYS2, ESRF, Grenoble, 20–22 May 2026.

2025

PLGrid computational grant: 500,000 core-hours on the Helios supercomputer

Grant no. PLG/2025/018524. Helios is operated by the Academic Computer Centre CYFRONET AGH at AGH University of Krakow, Kraków, Poland.

2022–present

Responsible for the implementation of the MITIQO project

Maintain and optimize the X-ray fluorescence setup and perform measurement campaigns. Expanded the scope of the Regione Lazio-funded project by implementing methods that improve sensitivity to trace metals (project no. A0375-2020-36647).

X-ray fluorescence analysis and protocol development

Investigate material composition, identify contaminants and quantify trace elements; develop protocols to improve the accuracy and reproducibility of XRF measurements.

Atomic spectroscopy

Analyse atomic spectra to identify and quantify elements and apply advanced spectroscopic techniques to study energy levels and transitions in atomic- and nuclear-physics research.

Machine learning

Develop and apply algorithms to analyse complex datasets and identify patterns; implement predictive models to improve research outcomes and optimize experimental procedures.

Member, INFN-LNF educational team

Contribute to activities for students at Liceo B. Touschek and to the “Incontri di Fisica” Winter Stages for high-school science teachers.

TECHNICAL EXPERTISE

X-ray methods	Laboratory XRF, high-resolution XES and developing XAS workflows; von Hamos geometry; Bragg crystals; X-ray tubes; motorized slits and stages; energy calibration; background subtraction and spectral fitting.
Detectors and analysis	CZT, BEGe, silicon PIN and position-sensitive detectors; detector-response modelling; spectral/waveform analysis; event classification; rate, efficiency, resolution and background studies.
Simulation	SHADOW4/OASYS2 ray tracing; Geant4 radiation transport; multi-configuration Dirac–Fock atomic calculations; atomic-cascade Monte Carlo; density-functional theory and high-throughput materials workflows.
Scientific computing	Python scientific computing; supervised and unsupervised machine learning; principal-component analysis; reproducible data-processing and simulation workflows; technical instruction and collaborative software coordination.

PUBLICATIONS MOST RELEVANT TO THE PROPOSAL

The five peer-reviewed articles below document the core competences supporting the proposal. The applicant's name is shown in bold; [*] denotes corresponding authorship and [†] equal contribution.

1. **S. Manti**[*] and A. Scordo, "Recent Developments of the VOXES Von Hamos X-Ray Spectrometer for Laboratory XES and XAS Studies," *Engineering Innovations* **19**, 23–29 (2026). doi:10.4028/p-rQpe9Z
2. **S. Manti**[*], F. Napolitano, A. Clozza, C. Curceanu, G. Moskal, K. Piscicchia, D. Sirghi and A. Scordo, "Enhancing Performances of the VOXES Bragg Spectrometer for XES Investigations," *Condensed Matter* **9**, 19 (2024). doi:10.3390/condmat9010019
3. **S. Manti**[*†], S.H. Yip[†] et al., "Machine-learning-enhanced event selection for BEGe detectors in the VIP experiment," *Journal of Instrumentation* **21**, P01036 (2026). doi:10.1088/1748-0221/21/01/P01036
4. **S. Manti**[*], F. Sgaramella, L. Abbene et al. (SIDDHARTA-2 Collaboration), "Precision test of bound-state QED at intermediate Z with kaonic neon," *Physical Review A* **113**, 022815 (2026). doi:10.1103/rry5-tdqb
5. K. Piscicchia, S. Donadi, **S. Manti**[*], A. Bassi, M. Derakhshani, L. Diósi and C. Curceanu, "X-Ray Emission from Atomic Systems Can Distinguish between Prevailing Dynamical Wave-Function Collapse Models," *Physical Review Letters* **132**, 250203 (2024). doi:10.1103/PhysRevLett.132.250203

ADDITIONAL SELECTED PUBLICATIONS AND PREPRINTS

6. F. Artibani, **S. Manti** et al., "Kaonic Copper and Fluorine Absolute Yields Measurement with a CZT-based Detection System at DAΦNE," preprint (2026). arXiv:2605.10477
7. **S. Manti**[*] et al., "Testing the Pauli Exclusion Principle across the Periodic Table with the VIP-3 Experiment," *Entropy* **26**, 752 (2024). doi:10.3390/e26090752
8. **S. Manti**[*], M.K. Svendsen, N.R. Knøsgaard, P.M. Lyngby and K.S. Thygesen, "Exploring and machine learning structural instabilities in 2D materials," *npj Computational Materials* **9**, 33 (2023). doi:10.1038/s41524-023-00977-x
9. F. Bertoldo[†], S. Ali[†], **S. Manti**[†] and K.S. Thygesen, "Quantum point defects in 2D materials – the QPOD database," *npj Computational Materials* **8**, 56 (2022). doi:10.1038/s41524-022-00730-w
10. S. Ali, F.A. Nilsson, **S. Manti**, F. Bertoldo, J.J. Mortensen and K.S. Thygesen, "High-Throughput Search for Triplet Point Defects with Narrow Emission Lines in 2D Materials," *ACS Nano* **17**, 21105–21115 (2023). doi:10.1021/acsnano.3c04774
11. **S. Manti**[*] et al., "Detecting Iron Oxidation States in Liquids with the VOXES Bragg Spectrometer," preprint (2023). arXiv:2310.06062

TEACHING, SUPERVISION AND OUTREACH

- Teaching Assistant, *Electronic Structure Methods in Materials Physics*, DTU, Spring 2019; and *Condensed Matter Physics and Nanoscale Materials Physics*, DTU, Fall 2018.
- Supervised students: Carlo Zamponi, Alex Tubby, Siu-Hung Yip, Michaela Lizardou and Luiz Gustavo Fernandes.
- M.Sc. thesis co-supervisor: Tobia Vecchiatto Montorio, University of Padua, *New Force Mediators and Their Impact on Exotic Atoms*, academic year 2025/2026.
- Contributor to INFN-LNF educational and outreach activities for the B. Touschek high-school programme and the "Incontri di Fisica" Winter Stages for science teachers.

PEER REVIEW

Reviewer for *Journal of Instrumentation (JINST)*, *Scientific Reports* and *Few-Body Systems*.

LANGUAGES

Italian	Native proficiency
English	Professional working proficiency
Danish	Elementary proficiency

SELECTED INVITED TALKS, LECTURES AND CONTRIBUTIONS

22–26 Jun. 2026 — **Lecturer, 2nd BridgeQG Training School**
"Testing quantum gravity with underground experiments."

14–19 Jun. 2026 — **Talk, European Conference on X-Ray Spectrometry (EXRS 2026), Catania**
"VOXES: a HAPG-Based Von Hamos Spectrometer for High-Resolution X-Ray Spectrometry of Extended Sources."

26 Mar. 2026 — **Oral contribution, XGRADE 2026, University of Palermo**
"Machine Learning background suppression for CZT detectors using Monte Carlo simulations."

3–5 Dec. 2025 — **Invited talk, Exotic Atoms: Fundamental Aspects, Applications and Advances in Radiation Detectors, Zagreb**
"Testing Bound-State QED in Strong Fields with Kaonic Atoms."

28 Oct.–1 Nov. 2025 — **Talk, 16th European Research Conference on Electromagnetic Interactions with Nucleons and Nuclei (EINN 2025), Paphos, Cyprus**
"AI-Enhanced BEGe Detectors for Low-Energy X-ray Collapse Model Tests."

8–10 Oct. 2025 — **Talk, Perspectives at the Bellotti IBF Workshop, LNGS, Assergi**
"Experiments at Bellotti IBF to Reach beyond the Standard Model."

24–26 Sep. 2025 — **Invited talk, 2nd Symposium on New Trends in Nuclear and Medical Physics, Krakow**
"High-Precision X-ray Spectroscopy: from Kaonic Atoms to Societal Applications."

24–29 Aug. 2025 — **Invited talk, Workshop on Standard Model and Beyond, Corfu**
"High-Precision Spectroscopy of Kaonic Atoms at SIDDHARTA-2: Probing Fundamental Interactions."

16–20 Jun. 2025 — **Talk and Conference Chair, High Precision X-Ray Measurements 2025, INFN-LNF**
"XRF Spectroscopy with VOXES: Techniques, Optimization and Applications."

30 Sep.–4 Oct. 2024 — **Talk, KAMPAI Workshop, ECT*, Trento**
"Kaonic Atom Properties and Cascade Models: Exploring Strong Interactions through First-Principles Calculations and Monte Carlo Simulations."

1–4 Jul. 2024 — **Invited talk, Advances in the Investigation of Weak and Strong Interactions, Bucharest, Romania**
"Kaonic Atom Properties and Cascade Models: Exploring Strong Interactions through First-Principles Calculations."

Jun. 2024 — **Invited talk, A Modern Odyssey: Quantum Gravity Meets Quantum Collapse, ECT*, Trento**
"Cancellation Effects in the Spontaneous Collapse Rate at Low-Energy Limit."

Nov. 2023 — **Invited talk, ALPACA Workshop, ECT*, Trento**
"Machine Learning in SIDDHARTA-2: the Monitoring Challenge."

19–23 Jun. 2023 — **Talk, High Precision X-Ray Measurements 2023, INFN-LNF**
"Investigating Metals' Content and Oxidation States in Edible Liquids using XRF Analysis with the VOXES X-Ray Spectrometer."

Oct. 2022 — **Invited talk, EXOTICO Workshop, ECT*, Trento**
"Cascade Models for Atomic Transitions in Kaonic Atoms."